



Quorum sensing inhibitory potential of vaccenic acid against *Chromobacterium violaceum* and methicillin-resistant *Staphylococcus aureus*

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Abstract

Quorum sensing (QS) is a potential target for inhibiting bacterial antibiotic resistance and associated pathogenicity. The present study aimed to investigate vaccenic acid anti-QS and antibiofilm potential against *Chromobacterium violaceum* and methicillin-resistant *Staphylococcus aureus* (MRSA). In the broth microdilution method, we determined the minimum inhibitory concentration (MIC) of vaccenic acid against *C. violaceum* and MRSA. Then, we determined the vaccenic acid anti-QS potential against *C. violaceum* via a violacein inhibition assay. Vaccenic acid at a sub-MIC concentration significantly inhibited violacein pigment production. Vaccenic acid also inhibits *C. violaceum* and MRSA biofilm formation at sub-MIC concentrations. The effect of vaccenic acid antivirulence potential was evaluated by phenotypic virulence assays. The results showed that vaccenic acid at a sub-MIC concentration significantly inhibited the virulence production of *C. violaceum* (chitinase and motility) and MRSA (hemolysin and staphyloxanthin production). Quantitative PCR analysis revealed the downregulation of QS associated genes upon vaccenic acid treatment. This resulted in the downregulation of genes involved in QS mechanisms such as *cviI*, *cviR*, and *SarA* and pigment production such as *vioB* and *crtM*. The results of the present study suggest that vaccenic acid is a promising agent to combat *C. violaceum* and MRSA.

Keywords *Staphylococcus aureus* · Vaccenic acid · *Chromobacterium violaceum* · *sarA* · Hemolysin · MRSA

Introduction

Obstruction of bacterial signaling communication is an alternative approach to restricting the antibiotic resistance associated with infections. Quorum sensing (QS) is a bacterial cell-to-cell communication system that is responsible for controlling bacterial behavior due to uncomfortable environments, and it activates virulence production, biofilm formation, virulence gene expression, motility, and pigment production (Pena et al. 2019). An autoinducer (AI) is a small chemical signaling molecule that initiates bacterial quorum sensing (QS). The autoinducer's production is

directly proportional to the bacterial cell density. Once they reach their secure concentration threshold; they contact a specific receptor protein and activate the various arrays of gene expression responsible for the formation of biofilm and virulence production (Rutherford and Bassler 2012). Generally, Gram-negative bacteria synthesize N-acylhomoserine lactones (AHLs) as AI-1 or AI-2 (Lu et al. 2005a, b), while at the same time, Gram-positive bacteria use the small oligopeptides as QS signal molecules (Verbeke et al. 2017).

Chromobacterium violaceum is a Gram-negative bacterium and is commonly used by biomarker strains in anti-quorum sensing experiments. *C. violaceum* synthesizes purple-colored pigment called violacein (Vasavi et al. 2013). A violacein pigment is formed when quorum sensing has succeeded. CviI/CviR regulates the *C. violaceum* QS system. It controlled phenotypic virulence factors such as biofilm formation, chitinolytic activity, elastase, and cyanide production (Stauff and Bassler 2011).

Methicillin-resistant *Staphylococcus aureus* (MRSA) is one of the dominant pathogenic bacteria commonly found in skin flora. This has the potential to result in hospital-acquired

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infections and increase the risk of patient mortality and morbidity. MRSA has the ability to cause numerous infections, such as skin infections, bacteremia, infective endocarditis, osteomyelitis, pneumonia, and burn wound infections (Tong et al. 2015; Hassoun et al. 2017; Lee et al. 2018). Previous studies have shown that MRSA infection rates are doubling in clinical settings due to the occurrence of multidrug resistance (MDR) (Klein et al. 2007; Kanerva et al. 2007; Köck et al. 2010). MRSA pathogenicity is associated with various virulence factors that protect the bacterial cells from the host immune system and antibiotics (Liu 2009). Biofilm formation is one of the major virulence factors that act as a primary barrier to protecting or preventing the bacterial cells from antibiotics (Roy et al. 2018). Adhesion proteins are capable of attaching to host tissues and organs. And enzymes associated with virulence, such as protease, hemolysis, and lipase, have the ability to digest and damage the host tissues (Cheung et al. 2021). Staphyloxanthin is a yellow colour pigment that acts as an antioxidant agent and is responsible for protecting the bacterial cells from oxidative stress (Clauditz et al. 2006). *S. aureus* pathogenicity is controlled by two major regulators, such as staphylococcal accessory regulator (SarA) and accessory gene regulator (agr). The virulence expression of *agr* and *sarA* has been extensively researched, revealing that *sarA* affects virulence gene expression via binding to the P2 and P3 promoter regions (Otto 2013). A previous study found that *sarA* regulates multiple adhesion genes through an *agr*-independent mechanism, and that a *sarA* mutant strain prevents the production of *S. aureus* biofilms. As a result, addressing *sarA* gene expression in clinical settings might be a way to control *S. aureus* pathogenesis.

Vaccenic acid is a naturally occurring trans-fatty acid that is commonly found in human milk. And dairy products such as milk, butter, and yogurt use vaccenic acid (Precht and Molkentin 1999). Short chain fatty acids such as acetate, propionate have been shown as quorum sensing quenchers (Lu et al. 2004), whereas limited scientific evidence exists regarding the effect of trans-fatty acids on antimicrobial potential, hence, the present study aimed to investigate the anti-quorum sensing and antibiofilm efficacy of trans fatty acids such as vaccenic acid against *C. violaceum* and MRSA and to reveal the mechanism underlying its anti-quorum sensing or antibiofilm potential.

Materials and methods

Bacterial strains and growth conditions

In this study, we used *Chromobacterium violaceum* (ATCC 12472) and methicillin resistant *Staphylococcus aureus* (ATCC 33591) for QSI assays. *C. violaceum* was cultured in Luria Bertani (LB) broth; MRSA was cultured in Brain

Heart Infusion (BHI) broth and incubated at 37 °C for 24 h. The stock cultures were preserved at – 80 °C in LB broth containing 20% glycerol.

Compound stock solution

Vaccenic acid was dissolved in DMSO (10 mg/ml) and stored at 4 °C for further use. Vaccenic acid was purchased from Sigma Aldrich (USA) with maximum purity. Methanol was used as vehicle control in all the assays where vaccenic acid was used.

Determination of minimum inhibitory concentration (MIC)

The minimum inhibitory concentration of vaccenic acid was determined using the broth microdilution method; the assay was performed in 24 well microtitre plates (CLSI 2018). Briefly, an overnight culture of *C. violaceum* and MRSA were diluted in LB or BHI broth to achieve a cell density of 10⁶ CFU/ml. A different concentration of vaccenic acid ranging from 16 to 1024 µg/ml was added to each well. Wells containing 1 ml of uninoculated broth or untreated bacterial culture served as negative and positive controls, respectively. The plates were incubated at 28 °C for *C. violaceum* and 37 °C for MRSA. After incubation, the plates were read at 600 nm.

Quantitative estimation of violacein inhibition

To determine quorum QSI potential of vaccenic acid against *C. violaceum*, the assay was performed in 24 well microtiter plate as described earlier (Vasavi et al. 2013). Briefly, an overnight culture of *C. violaceum* (1%) was mixed with 1 ml of LB broth and incubated in the presence and absence of vaccenic acid at sub-MIC concentration 1/8 (64 µg/ml), 1/4 (128 µg/ml) and 1/2 (256 µg/ml). The plate was incubated at 30 °C for 18 h. Then, the violacein pigment was quantified spectrophotometrically at OD 600 nm. After incubation, the plate was read at 600 nm.

Biofilm inhibition assay

The antibiofilm potential of vaccenic acid was assayed using 24 well microtiter plates as described in a previous study (Vijayakumar et al. 2020). *C. violaceum* and MRSA were cultured on LB broth and BHI broth for overnight and 1 ml of culture (adjusted to 10⁶ CFU/ml) were added into the wells. Subsequently, sub-MIC concentration of vaccenic acid 1/8 (64 µg/ml), 1/4 (128 µg/ml) and 1/2 (256 µg/ml) were added with the bacterial culture and PBS was used as positive and negative controls. The plates were incubated for 24 h at 28 °C for *C. violaceum* and 37 °C for MRSA.

After incubation, the plates were washed with sterile PBS to remove the unbound planktonic cells and allowed to dry for 1 min. For quantification, bound cells were stained with 0.4% of crystal violet (CV) for 10 min and excessive strains were removed by washing plates twice with sterile distilled water and air dried for 5 min. Then, biofilm cells were destained by using 20% of glacial acetic acid and OD was read at 570 nm.

Growth curve analysis

To evaluate the effect of QSI concentration of vaccenic acid on growth of *C. violaceum* and MRSA, 100 ml of LB or BHI was inoculated with an overnight culture of 1% of *C. violaceum* and MRSA in the presence and absence of 1/2 sub-MIC concentration of vaccenic acid. The zero hour OD was measured at 600 nm and the culture was incubated at 37 °C. The OD value was taken at 3 h interval for 24 h and the growth curve was plotted as OD against time interval (Vijayakumar and Muhilvannan 2021).

Chitinase inhibition assay

The effect of vaccenic acid on the chitinolytic activity of *C. violaceum*, the assay was performed by previous described method (Liu et al. 2013). The agar plates were prepared with supplementation of 0.4% K_2HPO_4 , 0.2% KH_2PO_4 , 0.15% $MgSO_4 \cdot 7H_2O$, 0.1% NaCl, 0.01% yeast extract, 1% chitin, and 1.5% bacteriological agar in the presence and absence of sub-MIC concentrations (1/2) of vaccenic acid. The plates were inoculated with an overnight culture of *C. violaceum* using a sterile toothpick and the plates were incubated at 28 °C for 72 h. The diameters of the transparent zone which formed around the *C. violaceum* colonies due to the breakdown of chitin were measured.

Swimming and swarming motility assay

To evaluate the effect of vaccenic acid on motility behavior in *C. violaceum* was assessed by previous described method (Yang et al. 2016). Briefly, LB broth supplemented with 0.5 agar was prepared and kept at 55 °C. Then, the sub-MIC (1/8 (64 µg/ml), 1/4 (128 µg/ml) and 1/2 (256 µg/ml) of vaccenic acid was added to the medium and vaccenic acid free medium was used as control. All the media were poured into petri dishes 2 h prior to use. A 5 µl of overnight culture of *C. violaceum* was used to inoculate plates by piercing the semisolid agar halfway.

In swarming motility assay, a semisolid agar plates prepared by above mentioned method except the agar was added at 0.5% and inoculation of *C. violaceum* were placed at the center of the agar surface. The plates were incubated

at 28 °C for 48 h and the diameters of the swimming and swarming zones were measured.

Hemolysin inhibition assay

MRSA was cultured in BHI broth containing sub-MIC (1/2) concentration of vaccenic acid and control BHI till they reached the post exponential phase (1×10^6 CFU/ml). Briefly, the culture supernatants were recovered after centrifugation at 5500 rpm at 4 °C. Aliquots of 100 µl of supernatant were brought up to 1 ml of hemolysin buffer solution (0.145 mol/L NaCl, 0.02 mol/L $CaCl_2$), then 25 µl of defibrinated sheep blood was added and incubated at 37 °C for 15 min. The liberated hemoglobin-containing supernatant was obtained by centrifugation at 5500 rpm at room temperature for 1 min and the absorbance was measured at 543 nm. The hemolytic activity of the vaccenic acid free supernatant was considered as 100% hemolysis. The percentage of hemolysis in presence of 1/2 (256 µg/ml) vaccenic acid was calculated as compared to the control (Xiang et al. 2010).

Staphyloxanthin inhibition assay

To evaluate the effect of vaccenic acid on carotenoid pigment production in MRSA was assessed by previous described method. Briefly, MRSA was grown in 100 ml of BHI broth along with sub-MIC (1/2) concentration of vaccenic acid at 37 °C for 24 h at 160 rpm. After incubation, the control and treated samples were centrifuged at 8000 rpm for 10 min, collected the pellets and washed twice with PBS. The pellets were taken for methanolic extraction by resuspending the pellets into methanol and kept in a shaker for 24 h. After incubation, the samples were centrifuged at 10,000 rpm for 10 min and the supernatant was measured at 465 nm (Leejae et al. 2013).

RNA extraction and cDNA synthesis

Total RNAs were extracted from the control and vaccenic acid (at 1/2 sub-MIC concentration) treated *C. violaceum* and MRSA cells using TRIzol method. Then, RNAs were converted to cDNA using High Capacity cDNA Reverse Transcription Kit (Applied Biosystems, USA).

Real time PCR analysis

Expression analysis was performed using real time PCR (7500 Sequence Detection System, Applied Biosystems, USA) for the candidate genes *cviI*, *cviR*, *vioB*, *sarA*, and *crtM* involved in QS mechanism, biofilm formation, and pigment production. PCR reaction was performed at predefined ratio using PCR Master Mix (SYBR Green kit, Applied Biosystems, USA). Expression pattern of candidate genes

was obtained by calculating $2^{-\Delta\Delta C_t}$ after normalizing the cycle threshold (C_t) values of respective genes against the C_t value of the housekeeping gene 16S rRNA (*C. violaceum*) and *gyrB* (MRSA). Primer sequences of candidate genes are given in Table 1.

Statistics

Each experiment was performed in triplicate to obtain statistical confidence. Data values of experimental results were recorded as the mean $\pm$ Standard Deviation. The significance was determined by using one way ANOVA as P -value < 0.05 . Statistical analyses were performed using SPSS 20.0 statistical software.

Results

The MIC of vaccenic acid on the growth of *C. violaceum* and MRSA

The result showed that vaccenic acid at increasing concentrations (16 $\mu\text{g/ml}$ to 1024 $\mu\text{g/ml}$) significantly inhibited bacterial growth. In particular, the 512 $\mu\text{g/ml}$ concentration exhibited the maximum inhibition. Hence, this concentration was considered by MIC. Then, a sub-MIC concentration of 512 $\mu\text{g/ml}$ such as 1/8 (64 $\mu\text{g/ml}$), 1/4 (128 $\mu\text{g/ml}$) and 1/2 (256 $\mu\text{g/ml}$) was used to evaluate the antibiofilm and anti-virulence agents against *C. violaceum* and MRSA.

Violacein inhibition assay

Vaccenic acid inhibited violacein production in a concentration-dependent manner. A 1/2 sub-MIC concentration of vaccenic acid inhibited 85% of violacein production, followed by 1/4 inhibited 68% and 1/8 inhibited 47%, respectively (Fig. 2b). Moreover, the results indicated that vaccenic acid has a quorum sensing inhibitory potential (QSI).

Biofilm inhibition assay

A vaccenic acid significantly inhibited biofilm formation in a concentration dependent manner (Fig. 3a). In *C. violaceum*, 1/2 (256 $\mu\text{g/ml}$) concentration showed a maximum inhibition (79%) followed by 1/4 (128 $\mu\text{g/ml}$) 35%, respectively (Fig. 3b). In MRSA, 1/2 (256 $\mu\text{g/ml}$) concentration showed a maximum inhibition (83%) followed by 1/4 (128 $\mu\text{g/ml}$) 40%, respectively (Fig. 3c).

Assay of the growth curve

The growth curve assay revealed that the antibiofilm activity of vaccenic acid is not related to its effect on bacterial growth. The OD of the bacterial suspensions treated with 1/2 sub-MIC (256 $\mu\text{g/ml}$) of vaccenic acid doesn't affect the bacterial growth compared to control samples (Fig. 4a, b).

Chitinase inhibition assay

The result showed that the diameter zone formation of chitin hydrolysis indicated that vaccenic acid significantly restricted the chitinase activity. Vaccenic acid at 1/2 (256 $\mu\text{g/ml}$) concentration significantly reduces the chitinase activity compared to control.

Inhibition of swimming and swarming motility

Vaccenic acid sub-MIC concentrations exhibited a significant inhibitory effect on both swimming and swarming motility in *C. violaceum*. In particular, the 1/2 (256 $\mu\text{g/ml}$) concentration significantly reduced the motility behavior compared to other concentrations. (Table 2).

MRSA virulence inhibition assay

Vaccenic acid at a sub-MIC (1/2) concentration significantly inhibited hemolysin production by 73% and staphyloxanthin production by 70%, respectively (Fig. 6).

Table 1 Gene specific primers used in this study

Gene	Forward primer	Reverse primer	Reference
16S rRNA	5'-GCGCAACCCCTTGCTCTAGT-3'	5'-TGTCACCGGCAGTCTCCTTA-3'	Liu et al. (2013)
<i>cvlI</i>	5'-CTGAAACTAAGCTGCGACAG-3'	5'-GAAACCGTCTCGCATAAG-3'	Liu et al. (2013)
<i>cvlR</i>	5'-CCCAGCAATATGCCGCTATC-3'	5'-CATTGAGCTTGCGATCACA-3'	Wang et al. (2019)
<i>vioB</i>	5'-ACGGTGGAGGAGGTGCGATTA-3'	5'-CGCACATCTGCCACATCAG-3'	Liu et al. (2013)
<i>agrA</i>	5'-TGATAATCCTTATGAGGTGC-3'	5'-CACTGTGACTGGTAACGAAA-3'	Vijayakumar et al. (2020)
<i>sarA</i>	5'-CAAACAACCACAAGTTGTTA-3'	5'-TGTTTGCTTCAGTGATTCGT-3'	Vijayakumar et al. (2020)
<i>crtM</i>	5'-ATCCAGAACCACCCGTTTT-3'	5'-GCGATGAAGGTATTGGCAT-3'	Vijayakumar et al. (2020)
<i>gyrB</i>	5'-GGTGCTGGGCAAATACAAGT-3'	5'-TCCACACTAAATGCTGCAA-3'	Vijayakumar et al. (2020)

Table 2 Effect of vaccenic acid at sub-MIC concentration on swimming and swarming motility of *C.violaceum*

Assays	Radius migration zone after 48 h (mm)			
	Control	1/8	1/4	1/2
Swimming assay	38.2±0.5	29.4±0.8	15.6±1.2	8.8±0.5
Swarming assay	27.3±1.8	19.4±0.5	12.2±1.9	8.3±1.7

Quantitative expression of QS regulated genes

RT-PCR results revealed that vaccenic acid treatment (1/2 sub-MIC concentration) significantly inhibited the expression of *cviI* (1.5), *cviR* (2.1) and *vioB* (2.5) genes in *C. violaceum* (Fig. 7). In MRSA, vaccenic acid treatment significantly downregulated the expression of *agrA* (3.1), *sarA* (2.4) and *crtM* (1.7) genes (Fig. 8), respectively.

Discussion

The antibiotic resistance emergency imposes the search for alternative approaches to battle against bacterial infections. The present study demonstrates vaccenic acid as a new promising source of anti-infective agents that could be used

to inhibit QS-mediated bacterial pathogenicity. To the best of our knowledge, this is the first report on the QSI effect of vaccenic acid. First, we determined the MIC value of vaccenic acid against the target bacteria. Vaccenic acid affected bacterial growth at 512 µg/ml concentration (Fig. 1a, b). Then, we quantified the violacein pigment production of *C. violaceum* when treated with a sub-MIC concentration of vaccenic acid. Vaccenic acid significantly reduced violacein pigment production in a concentration-dependent manner (Fig. 2a, b). Violacein pigment production is controlled by QS in *C. violaceum*. The *C. violaceum* QS system begins with the response regulator. *cviR* is a response regulator for the CviI/CviR mediated QS system and it positively regulates the promoter genes like *vioA*, *vioB*, *vioC*, *vioD*, and *vioE*, respectively. The operon system *vioABCDE* plays a key role in violacein production (Kothari et al. 2017). Hence, the inhibitory effect of vaccenic acid on violacein production perceived in this study could be recognized as a QSI potential on the CviI/CviR system. The present finding was compared with a previous study in which bergamot and aspidosperma were able to inhibit violacein pigment production at sub-MIC concentrations (Ahmed et al. 2021).

Biofilm is a key virulence factor functionalized under QS. A biofilm has been formed on biotic or abiotic surfaces, including medicinal devices. Four steps, such as initial

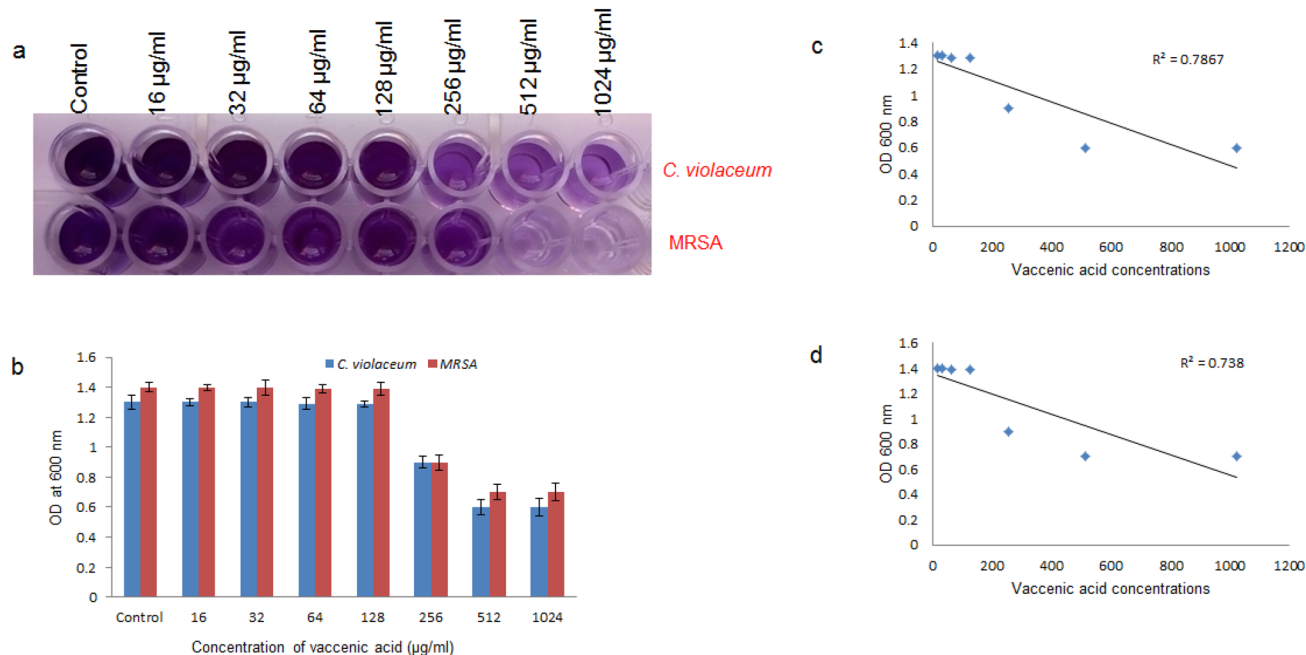


Fig. 1 **a** Antibacterial activity of vaccenic acid against *C. violaceum* and MRSA. 96-well plate exposed concentration-dependent inhibition of *C. violaceum* and MRSA bacterial growth. **b** Graph showing percentage inhibition of *C. violaceum* and MRSA bacterial growth by vaccenic acid in a concentration-dependent manner. A 512 µg/ml concentration inhibited both bacterial growths. Hence, this concentra-

tion was chosen as MIC. A further study has to be done by sub-MIC concentrations of 516 µg/ml (1/8 (64 µg/ml, 1/4 (128 µg/ml and 1/2 (256 µg/ml). Error bars indicates SDs from the mean [$n=6$ (biological triplicates in technical duplicates)]. * indicate a statistically significant difference ($p<0.05$). The dose and response correlation was analyzed and the R2 value is 0.78 (*C. violaceum*) and 0.73 (MRSA)

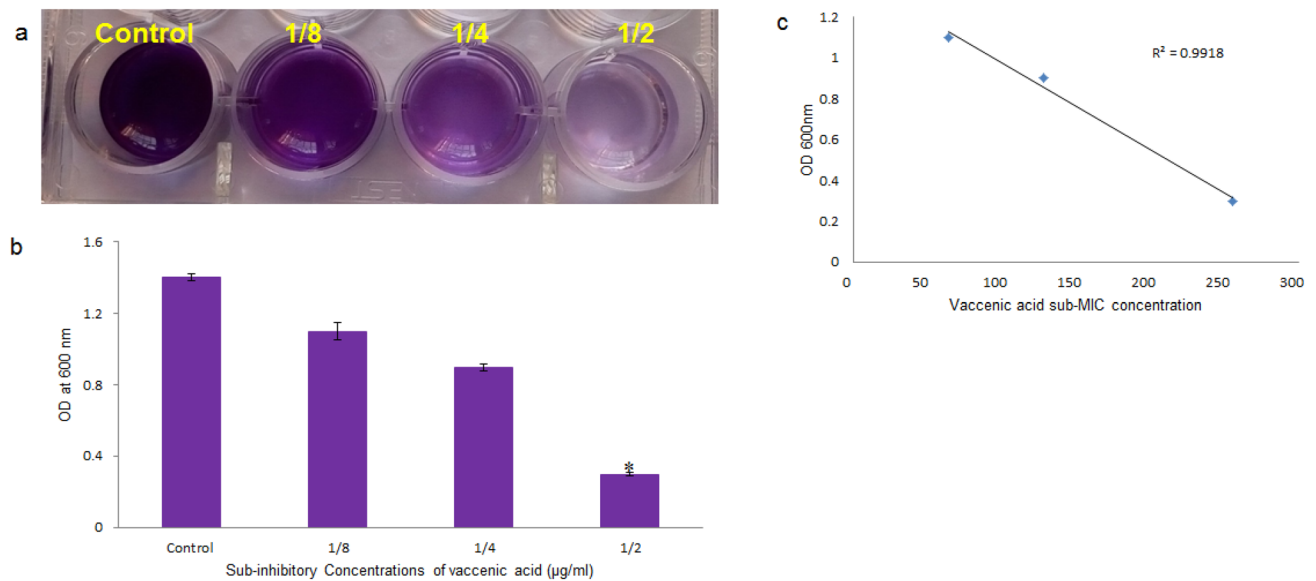


Fig. 2 **a** Effect of vaccenic acid on violacein pigment production of *C. violaceum*. **%**. A 24-well plate exposed concentration-dependent inhibition of *C. violaceum* pigment production. **b** Graph showed percentage inhibition of *C. violaceum* pigment production by vaccenic

acid in a concentration-dependent manner. Error bars indicates SDs from the mean [$n=6$ (biological triplicates in technical duplicates)]. * indicate a statistically significant difference ($p < 0.05$). The dose and response correlation was analyzed and the R^2 value is 0.99

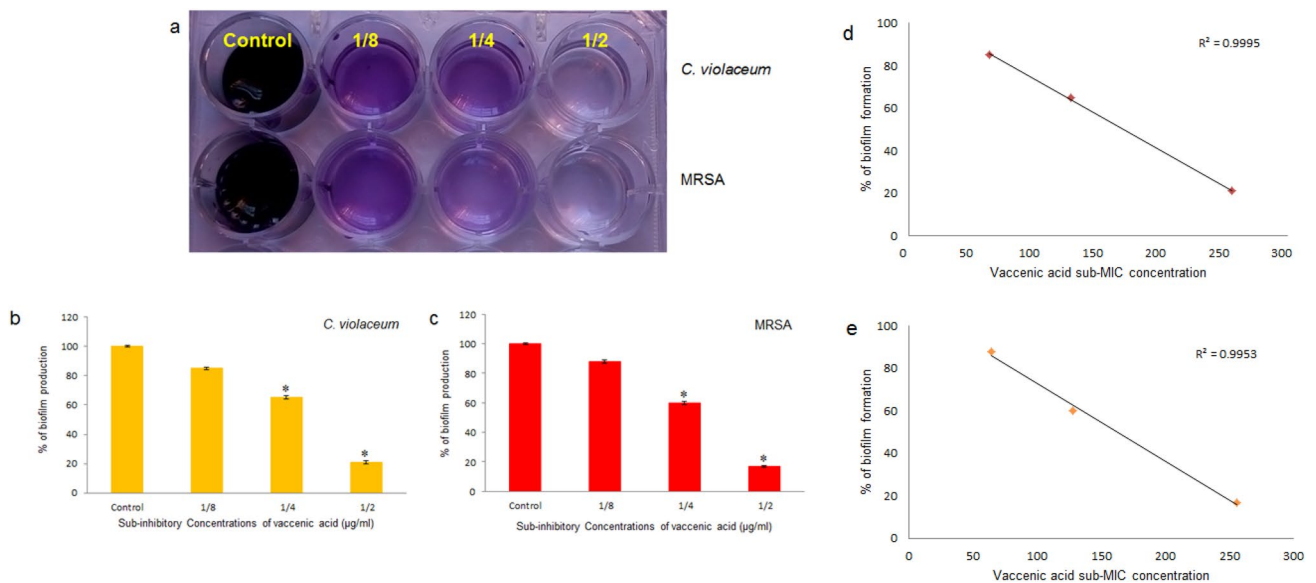


Fig. 3 **a** Antibiofilm potential of vaccenic acid against *C. violaceum* and MRSA. 24-well plate exposed concentration-dependent inhibition of *C. violaceum* and MRSA biofilm formation. **b** Graph showing percentage inhibition of *C. violaceum* biofilm formation by vaccenic acid. A 1/2 (256 μg/ml) sub-MIC concentration showed a maximum reduction of biofilm formation by 87%. **c** Graph showing percentage inhibition of MRSA biofilm formation by vaccenic acid. A 1/2 (256 μg/

ml) sub-MIC concentration showed a maximum reduction of biofilm formation by 82%. Then, a further study has to be done by this concentration. Error bars indicates SDs from the mean [$n=6$ (biological triplicates in technical duplicates)]. * indicate a statistically significant difference ($p < 0.05$). The dose and response correlation was analyzed and the R^2 value is 0.99 (*C. violaceum*) and 0.99 (MRSA)

attachment, colony formation, maturation, and dispersion, are needed to complete the entire biofilm structure (Otto 2013). Once a biofilm has developed, they become resistant

to antibiotics. The biofilms default to protecting the bacterial cells from lack of nutrients, environmental stresses, antibiotics, and host immune response (Khatoon et al. 2018).

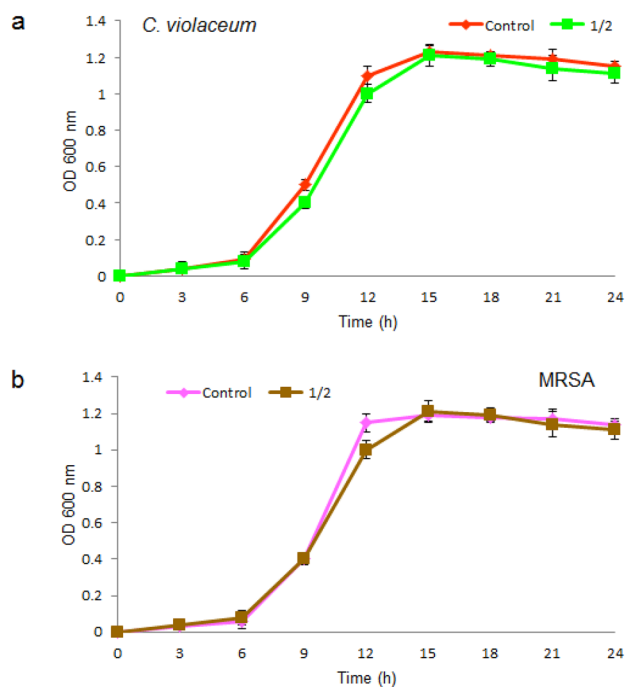


Fig. 4 Effect of vaccenic acid sub-MIC concentration on bacterial growth of *C. violaceum* and MRSA. **a** Vaccenic acid at 1/2 (256 µg/ml) sub-MIC concentration doesn't affect the bacterial growth of *C. violaceum* compared to control. **b** Vaccenic acid at 1/2 (256 µg/ml) sub-MIC concentration doesn't affect the bacterial growth of MRSA compared to control. Error bars indicates SDs from the mean [$n=6$ (biological triplicates in technical duplicates)]

Hence, biofilm is the main target for controlling biofilm-associated bacterial pathogenicity. In the present study, we evaluated vaccenic acid for its ability to reduce biofilm formation by *C. violaceum* and MRSA. Vaccenic acid greatly reduced biofilm formation by *C. violaceum* and MRSA at a sub-MIC concentration (Fig. 3a–c). At the same time, vaccenic acid sub-MIC concentration does not possess any bactericidal effect against *C. violaceum* and MRSA (Fig. 4a, b). The results indicate that vaccenic acid is a promising anti-infective agent. The current finding was compared to a previous study, which found that combining tannic acid with antibiotics significantly reduced *C. violaceum* biofilm formation (Dusane et al. 2015). In a recent study assessing phytochemicals, myrtenol reported antibiofilm activity against MRSA (Selvaraj et al. 2019).

The sub-MIC concentration of vaccenic acid showed no significant effect on bacterial growth; hence, this concentration was selected an investigating the antivirulence effect of vaccenic acid. Then, we evaluated the antivirulence potential of vaccenic acid on phenotypic virulence factors. Chitinase is an enzyme that allows *C. violaceum* growth on chitin as a carbon source (Becker et al. 2009). Duran et al. (2016) reported that CviR has been directly activating chitinase production (Duran et al. 2016). This data confirmed that the

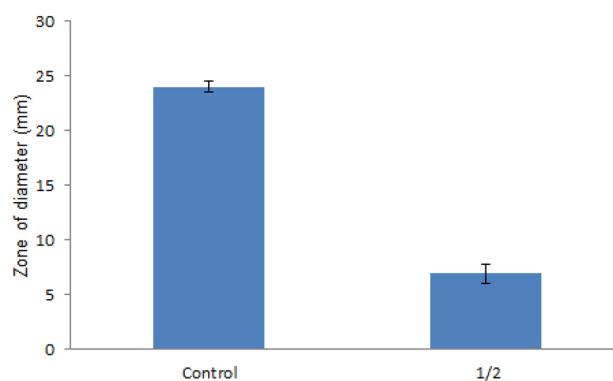


Fig. 5 Inhibition of chitinolytic activity of *C. violaceum*. The effect of sub-MIC concentration 1/2 (256 µg/ml) of vaccenic acid on chitinase production by *C. violaceum* was tested on chitin plates. After the incubation, chitin hydrolysis diameters zones were measured. Error bars indicates SDs from the mean [$n=6$ (biological triplicates in technical duplicates)]

C. violaceum QS is responsible for chitinase production. In the present study, it was revealed that chitinase production was inhibited at a sub-MIC concentration of vaccenic acid treatment (Fig. 5). The result indicated the inhibitory effect of vaccenic acid on the CviI/R system. The present finding was compared to a prior study that found kitasamycin and nitrofurantoin reduced *C. violaceum* chitinase synthesis at sub-MIC levels (Ibrahim et al. 2020).

Bacterial motility is responsible for nutrient attainment, colonization, and lethal environment evasion, and is involved in the spread of bacteria that invade the host (McDougald et al. 2012). Swarming is a multicellular type of motility controlled by the QS system, and it allows bacteria to proliferate on semi-solid surfaces. A previous study reported that swarming is quite often linked with biofilm formation and has been associated with pathogenic virulence factors (Bindel Connelly et al. 2004). Swarming motility is recognized as one of the virulence factors in Gram negative bacteria, in particular, *C. violaceum*, and it has been associated with bacterial pathogenicity and antibiotic resistance (Josenhans and Suerbaun 2002). Swimming motility is a flagella-dependent individual cell movement in broth or semi-solid medium (Kearns 2010). The present study revealed that *C. violaceum* was diminished in both swimming and swarming motilities upon being treated with vaccenic acid (Table. 2). The present finding indicates that vaccenic acid interrupts QS-mediated motility and flagella-mediated motility in *C. violaceum*. The result was compared with previous studies in which tannic acid was able to inhibit swimming motility (Dusane et al. 2015) and *Cinnamomum camphora* leaf essential oil inhibits swarming motility (Wang et al. 2019) in *C. violaceum*.

MRSA is a well-recognized QS-associated pathogen. Hence, we aimed to evaluate the effect of vaccenic acid

on the production of virulence factors in MRSA. Vaccenic acid at a sub-MIC concentration (1/2 MIC) significantly suppressed the hemolytic and staphyloxanthin production in MRSA (Fig. 6). Hemolysin is one of the important β -barrel pores forming exotoxins that are responsible for the lysis of numerous host cells such as erythrocytes, macrophages, endothelial cells, epithelial cells, and monocytes, and it is also involved in *S. aureus* infections such as septicemia, pneumonia, and skin infections (Berube and Wardenburg 2013; Chua et al. 2014; Tavares et al. 2014). Antioxidant production is required for MRSA growth and survival against reactive oxygen species (ROS). Staphyloxanthin is an antioxidant and a golden-yellow colored carotenoid pigment that is responsible for MRSA resistance to ROS (Clauditz et al. 2006). In this study, vaccenic acid (at 1/2 sub-MIC concentration) treatment significantly inhibited staphyloxanthin pigment production. The obtained results were similar to the previous study. Abbas et al. (2020) reported that the sub-MIC concentration (62.5 $\mu\text{g/ml}$) of diclofenac inhibited the hemolysin activity and staphyloxanthin production in MRSA (Abbas et al. 2020).

Further, vaccenic acid QSI potential was evaluated at the molecular level by using the RT-qPCR method. *C. violaceum* and MRSA are well-known QS-producing bacteria. They regulate the expression of virulence genes based on bacterial cell population density. In *C. violaceum*, pigment production is controlled by the CviI/CviR QS system. *cviI* gene acts as an autoinducer, which encodes the signal AHL, and *cviR* is a receptor that regulates virulence production, including violacein in *C. violaceum*. QS is a cell density based system, where cell density increases are directly proportional to the increase in AHL production. When the cell density reaches a certain threshold, *cviI* synthesises and releases the AHL with a stable protein-ligand complex, which is responsible for finding a binding region called the receptor (*cviR*) and activating the *vio* operon system. A *vio* operon consists of five enzyme genes called *vioA*, *vioB*, *vioC*, *vioD*, and *vioE* that are transcribed in one direction (Duran et al. 2016). Following that, the AHL: CviR complex binds to the promoter region of the *vio* operon and activates the production of violacein pigment (Kothari et al. 2017). The present study revealed that vaccenic acid at a sub-MIC concentration significantly downregulated the *cviI* and *vioB* genes (Fig. 7). The present findings suggest that the QSI potential of vaccenic acid could affect the AHL signal production or interrupt AHL: CviR complex binding to the *vio* operon promoter site. In a previous study, Duran et al. (2016) reported that CviR positively regulated the chitinase production, type VI secretion system, and *cviI* gene expression in *C. violaceum* (Duran et al. 2016). Hence, the downregulation of *cviI* could be a possible reason for inhibiting virulence production such as biofilm formation, chitinase production, and motility in *C. violaceum* upon vaccenic acid treatment

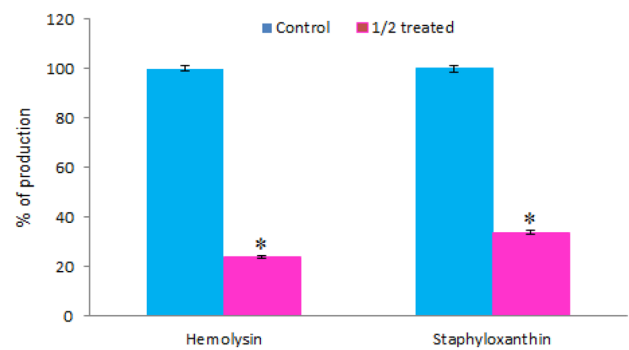


Fig. 6 Effect of sub-MIC concentration 1/2 (256 $\mu\text{g/ml}$) of vaccenic acid inhibits MRSA virulence factors such as hemolysin and staphyloxanthin production. Error bars indicates SDs from the mean [$n=6$ (biological triplicates in technical duplicates)]. * indicate a statistically significant difference ($p < 0.05$)

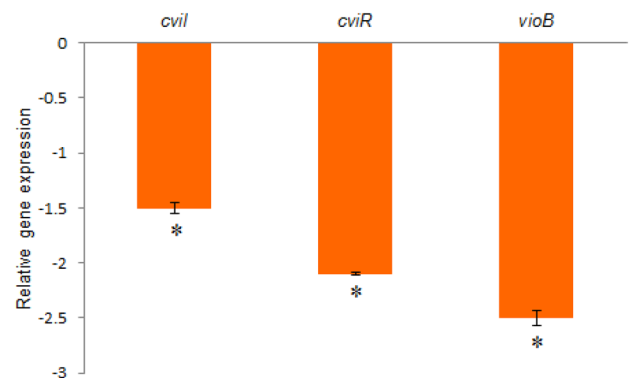


Fig. 7 Effect of vaccenic acid on the expression of QS genes. Relative gene expression was measured by RT-qPCR for the *cviI*, *cviR* and *vioB* genes of *C. violaceum*. Error bars indicates SDs from the mean [$n=6$ (biological triplicates in technical duplicates)]. * indicate a statistically significant difference ($p < 0.05$)

(Fig. 7). The present findings are similar to the previous study in which Ahmed and his team reduced the *C. violaceum* virulence production due to the suppression of *cviI* and *vioB* genes. (Ahmed et al. 2021).

On the other hand, the effect of vaccenic acid was assessed on regulator genes and virulence gene expression of MRSA. The two-component regulatory system (TCRS) is well associated with the QS system in *S. aureus*. Agr (accessory gene regulator) is a well-studied TCRS in *S. aureus*. It regulates the numerous virulence products. Agr is encoded by the agr locus and consists of two different transcriptional units, such as RNAII and RNAPIII, which are regulated by P2 and P3 promoters (Novick 2003). The Agr QS system is controlled by the RNAII transcript and contains four groups of genes, including *AgrBDCA*. AgrD is a precursor peptide of AIP, AgrB is involved in the development of mature AIP, and AgrC is a two-component sensor histidine kinase. When AIP binds to AgrC, this

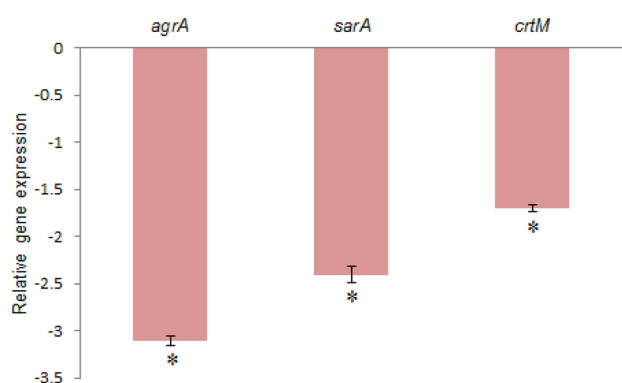


Fig. 8 Effect of vaccenic acid on the expression of QS genes. Relative gene expression was measured by RT-qPCR for the *sarA* and *crtM* genes of MRSA. Error bars indicates SDs from the mean [$n=6$ (biological triplicates in technical duplicates)]. * indicate a statistically significant difference ($p < 0.05$)

reaction transforms the phosphate from AgrC to AgrA, and AgrA activates the P2 and P3 promoters (Yarwood et al. 2004; Eickhoff and Bassler 2018). In the present study, vaccenic acid treatment significantly downregulated *agrA* gene expression (Fig. 8). The current data shows that the vaccenic acid QSI effect can be classified as Agr locus inhibition. The results were consistent with a previous study on the QSI potential of diclofenac in inhibiting AgrA expression in MRSA (Abbas et al. 2020). *SarA* (staphylococcal accessory regulator A) is another master regulator whose response is to regulate extracellular proteins, in particular, α -hemolysin (Gordon et al. 2013). Vaccenic acid treatment downregulated the expression of *sarA* (Fig. 8) and this could be a possible reason behind the antibiofilm and antihemolytic activity of vaccenic acid, as a previous study reported that *sarA* mediated antibiofilm potential against *S. aureus* via suppression of *sarA* expression (Valliammai et al. 2019). Staphyloxanthin is a carotenoid pigment that is responsible for maintaining the antioxidant level in *S. aureus*. In *S. aureus*, *crtM* (4,4'-diapophytoene synthase) and *crtN* (4, 4'-diapophytoene desaturase) encoded staphyloxanthin production, which was studied, and it was found that vaccenic acid significantly downregulated the expression of *crtM* (Fig. 8) which could be a possible reason for the inhibition of staphyloxanthin production in the virulence assay.

In conclusion, the present study demonstrated vaccenic acid as a promising source of QSI agents. Vaccenic acid has inhibitory potential against QS and mediates virulence factors of *C. violaceum* and MRSA, including biofilm formation, violacein production, chitinase production, anti-motility, hemolysin production, and staphyloxanthin production. As a result, vaccenic acid may be equipped to treat or control the bacterial infections via regulating QS. However, proteomic analysis and animal experiments are required.

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Declarations

Conflict of interest All the authors declare that they have no conflict of interest.

Ethical approval This study does not contain any experiments with human participants or animals performed by any of the authors.

References

- Abbas HA, Atallah H, El-Sayed MA, El-Ganiny AM (2020) Diclofenac mitigates virulence of multidrug-resistant *Staphylococcus aureus*. Arch Microbiol 202:2751–2760
- Ahmed SO, Zedan HH, Ibrahim YM (2021) Quorum sensing inhibitory effect of bergamot oil and aspidosperma extract against *Chromobacterium violaceum* and *Pseudomonas aeruginosa*. Arch Microbiol 203:4663–4675
- Becker S, Soares C, Porto LM (2009) Computational analysis suggests that virulence of *Chromobacterium violaceum* might be linked to biofilm formation and poly-NAG biosynthesis. Genet Mol Biol 32:640–644
- Berube BJ, Wardenburg JB (2013) *Staphylococcus aureus* α -toxin: nearly a century of intrigue. Toxins 5:1140–1166
- Bindel Connelly M, Young GM, Sloma A (2004) Extracellular proteolytic activity plays a central role in swarming motility in *Bacillus subtilis*. J Bacteriol 186:4159–4167
- Cheung GY, Bae JS, Otto M (2021) Pathogenicity and virulence of *Staphylococcus aureus*. Virulence 12:547–569
- Chua KY, Monk IR, Lin YH, Seemann T, Tuck KL, Porter JL, Stepanell J, Coombs GW, Davies JK, Stinear TP, Howden BP (2014) Hyperexpression of α -hemolysin explains enhanced virulence of sequence type 93 community-associated methicillin-resistant *Staphylococcus aureus*. BMC Microbiol 14:1–13
- Clauditz A, Resch A, Wieland KP, Peschel A, Götz F (2006) Staphyloxanthin plays a role in the fitness of *Staphylococcus aureus* and its ability to cope with oxidative stress. Infect Immun 74:4950–4953
- Durán N, Justo GZ, Durán M, Brocchi M, Cordi L, Tasic L, Castro GR, Nakazato G (2016) Advances in *Chromobacterium violaceum* and properties of violacein-Its main secondary metabolite: a review. Biotechnol Adv 34:1030–1045
- Dusane DH, O'May C, Tufenkji N (2015) Effect of tannic and gallic acids alone or in combination with carbenicillin or tetracycline on *Chromobacterium violaceum* CV026 growth, motility, and biofilm formation. Can J Microbiol 61:487–494
- Eickhoff MJ, Bassler BL (2018) SnapShot: bacterial quorum sensing. Cell 174:1328–1328
- Gordon CP, Williams P, Chan WC (2013) Attenuating *Staphylococcus aureus* virulence gene regulation: a medicinal chemistry perspective. J Med Chem 56:1389–1404
- Hassoun A, Linden PK, Friedman B (2017) Incidence, prevalence, and management of MRSA bacteremia across patient populations: a

- review of recent developments in MRSA management and treatment. *Crit Care* 21:1
- Ibrahim YM, Abouwarda AM, Omar FA (2020) Effect of kitasamycin and nitrofurantoin at subinhibitory concentrations on quorum sensing regulated traits of *Chromobacterium violaceum*. *Antonie Van Leeuwenhoek* 113:1601–1615
- Josenhans C, Suerbaum S (2002) The role of motility as a virulence factor in bacteria. *Int J Med Microbiol* 291:605–614
- Kanerva M, Blom M, Tuominen U, Kolho E, Anttila VJ, Vaara M, Virolainen-Julkunen A, Lyytikäinen O (2007) Costs of an outbreak of methicillin-resistant *Staphylococcus aureus*. *J Hosp Infect* 66:22–28
- Kearns DB (2010) A field guide to bacterial swarming motility. *Nat Rev Microbiol* 8:634–644
- Khatoun Z, McTiernan CD, Suuronen EJ, Mah TF, Alarcon EI (2018) Bacterial biofilm formation on implantable devices and approaches to its treatment and prevention. *Heliyon* 4:e01067
- Klein E, Smith DL, Laxminarayan R (2007) Hospitalizations and deaths caused by methicillin-resistant *Staphylococcus aureus*, United States, 1999–2005. *Emerg Infect Dis* 13:1840
- Köck R, Becker K, Cookson B, van Gemert-Pijnen JE, Harbarth S, Kluytmans JA, Mielke M, Peters G, Skov RL, Struelens MJ, Tacconelli E (2010) Methicillin-resistant *Staphylococcus aureus* (MRSA): burden of disease and control challenges in Europe. *Eurosurveillance* 15:19688
- Kothari V, Sharma S, Padia D (2017) Recent research advances on *Chromobacterium violaceum*. *Asian Pac J Trop Med* 10:744–752
- Lee AS, de Lencastre H, Garau J, Kluytmans J, Malhotra-Kumar S, Peschel A, Harbarth S (2018) Methicillin-resistant *Staphylococcus aureus*. *Nat Rev Dis Primers* 4:1–23
- Leejae S, Hasap L, Voravuthikunchai SP (2013) Inhibition of staphyloxanthin biosynthesis in *Staphylococcus aureus* by rhodomycinone, a novel antibiotic candidate. *J Med Microbiol* 62:421–428
- Liu GY (2009) Molecular pathogenesis of *Staphylococcus aureus* infection. *Pediatr Res* 65:71–77
- Liu Z, Wang W, Zhu Y, Gong Q, Yu W, Lu X (2013) Antibiotics at sub-inhibitory concentrations improve the quorum sensing behavior of *Chromobacterium violaceum*. *FEMS Microbiol Lett* 341:37–44
- Lu L, Hume ME, Pillai SD (2004) Autoinducer-2-like activity associated with foods and its interaction with food additives. *J Food Prot* 67:1457–1462
- Lu L, Hume ME, Pillai, (2005a) Autoinducer 2-like activity in poultry-associated enteric bacteria in response to subtherapeutic antibiotic exposure. *Avian Dis* 49:74–80
- Lu L, Hume ME, Pillai SD (2005b) Autoinducer-2-like activity on vegetable produce and its potential involvement in bacterial biofilm formation on tomatoes. *Foodborne Pathog Dis* 2:242–249
- McDougald D, Rice SA, Barraud N, Steinberg PD, Kjelleberg S (2012) Should we stay or should we go: mechanisms and ecological consequences for biofilm dispersal. *Nat Rev Microbiol* 10:39–50
- Novick RP (2003) Autoinduction and signal transduction in the regulation of staphylococcal virulence. *Mol Microbiol* 48:1429–1449
- Otto M (2013) Staphylococcal infections: mechanisms of biofilm maturation and detachment as critical determinants of pathogenicity. *Annu Rev Med* 64:175–188
- Pena RT, Blasco L, Ambroa A, González-Pedraja B, Fernández-García L, López M, Bleriot I, Bou G, García-Contreras R, Wood TK, Tomás M (2019) Relationship between quorum sensing and secretion systems. *Front Microbiol* 10:1100
- Precht D, Molkenin J (1999) C18: 1, C18: 2 and C18: 3 trans and cis fatty acid isomers including conjugated cis $\Delta 9$, trans $\Delta 11$ linoleic acid (CLA) as well as total fat composition of German human milk lipids. *Food Nahrung* 43:233–244
- Roy R, Tiwari M, Donelli G, Tiwari V (2018) Strategies for combating bacterial biofilms: a focus on antibiofilm agents and their mechanisms of action. *Virulence* 9:522–554
- Rutherford ST, Bassler BL (2012) Bacterial quorum sensing: its role in virulence and possibilities for its control. *Cold Spring Harb Perspect Med* 2:a012427
- Selvaraj A, Jayasree T, Valliammai A, Pandian SK (2019) Myrtenol attenuates MRSA biofilm and virulence by suppressing sarA expression dynamism. *Front Microbiol* 2019:2027
- Stauff DL, Bassler BL (2011) Quorum sensing in *Chromobacterium violaceum*: DNA recognition and gene regulation by the CviR receptor. *J Bacteriol* 193:3871–3878
- Tavares A, Nielsen JB, Boye K, Rohde S, Paulo AC, Westh H, Schønning K, de Lencastre H, Miragaia M (2014) Insights into alpha-hemolysin (Hla) evolution and expression among *Staphylococcus aureus* clones with hospital and community origin. *PLoS ONE* 9:e98634
- Tong SY, Davis JS, Eichenberger E, Holland TL, Fowler VG Jr (2015) *Staphylococcus aureus* infections: epidemiology, pathophysiology, clinical manifestations, and management. *Clin Microbiol Rev* 28:603–661
- Valliammai A, Sethupathy S, Priya A, Selvaraj A, Bhaskar JP, Krishnan V, Pandian SK (2019) 5-Dodecanolide interferes with biofilm formation and reduces the virulence of Methicillin-resistant *Staphylococcus aureus* (MRSA) through up regulation of agr system. *Sci Rep* 9:1–16
- Vasavi HS, Arun AB, Rekha PD (2013) Inhibition of quorum sensing in *Chromobacterium violaceum* by *Syzygium cumini* L. and *Pimenta dioica* L. *Asian Pac J Trop Biomed* 3:954–959
- Verbeke F, De Craemer S, Debonne N, Janssens Y, Wynendaele E, Van de Wiele C, De Spiegeleer B (2017) Peptides as quorum sensing molecules: measurement techniques and obtained levels in vitro and in vivo. *Front Neurosci* 11:183
- Vijayakumar K, MuhilVannan S (2021) 3, 5-Di-tert-butylphenol combat against *Streptococcus mutans* by impeding acidogenicity, acidurance and biofilm formation. *World J Microbiol Biotechnol* 37:1–13
- Vijayakumar K, Bharathidasan V, Manigandan V, Jeyapragash D (2020) Quebrachitol inhibits biofilm formation and virulence production against methicillin-resistant *Staphylococcus aureus*. *Microb Pathog* 149:104286
- Wang W, Li D, Huang X, Yang H, Qiu Z, Zou L, Liang Q, Shi Y, Wu Y, Wu S, Yang C (2019) Study on antibacterial and quorum-sensing inhibition activities of *Cinnamomum camphora* leaf essential oil. *Molecules* 24:3792
- Wayne, PA (2018) Clinical and laboratory standards institute performance standards for antimicrobial susceptibility testing. Clinical and Laboratory Standards Institute.
- Xiang H, Qiu JZ, Wang DC, Jiang YS, Xia LJ, Deng XM (2010) Influence of Magnolol on the Secretion of α -Toxin by *Staphylococcus aureus*. *Molecules* 15:1679–1689
- Yang Q, Wang L, Gao J, Liu X, Feng Y, Wu Q, Baloch AB, Cui L, Xia X (2016) Tannin-rich fraction from pomegranate rind inhibits quorum sensing in *Chromobacterium violaceum* and biofilm formation in *Escherichia coli*. *Foodborne Pathog Dis* 13:28–35
- Yarwood JM, Bartels DJ, Volper EM, Greenberg EP (2004) Quorum sensing in *Staphylococcus aureus* biofilms. *J Bacteriol* 186:1838–1850

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